Software Requirements Specification (SRS)

PrivacyGuard

Project: PrivacyGuard

Platform: Android

Main Programming Language: Java

Development Environment: Android Studio

Team Size: 3 students

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1. Introduction

1.1 Purpose

PrivacyGuard is an Android mobile application designed to help users identify personal information and potential privacy risks in their content before sharing it online.

The purpose of this SRS is to describe the main functions and requirements of the PrivacyGuard application.

1.2 Scope

PrivacyGuard will allow users to analyze text, images, and screenshots for potentially sensitive information such as phone numbers, email addresses, names, addresses, locations, school/workplace information, and identification information.

The application will also check relevant image metadata, such as GPS information, when available.

After analyzing the content, the application will provide a risk score and risk level, explain the detected risks, and give recommendations to help the user reduce their exposure before sharing the content.

The application will be developed for Android using Java as the main programming language. Other tools and libraries may be used to support functions such as OCR and image analysis.

The current version will not include Terms & Conditions analysis, direct social-media integration, automatic posting, or continuous monitoring of the user's device.

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2. Overall Description

2.1 Product Perspective

PrivacyGuard is a standalone Android application. The user provides content to the application, and the application analyzes it and displays the detected privacy risks.

The main workflow is:

Select/Input Content → Analyze Content → Detect Personal Information → Assess Risk → Display Results → Give Recommendations

The main technologies are:

- Java for the application logic
- Android SDK for mobile functionality
- Android Studio for development
- OCR technology for extracting text from images
- Image metadata analysis for detecting information such as GPS coordinates
- Rule-based or other suitable detection methods for identifying personal information

Python may be used by the team during research or experimentation, but it is not a required component of the final Android application.

2.2 Product Functions

The main functions of PrivacyGuard are:

1. Allow the user to enter text for analysis.
2. Allow the user to select an image or screenshot from their device.
3. Analyze text contained in images using OCR.
4. Detect potentially sensitive personal information.
5. Check available image metadata.
6. Identify possible location information, including GPS metadata.
7. Calculate an overall privacy risk level.
8. Display the detected risks to the user.
9. Explain why the detected information may represent a privacy risk.
10. Provide recommendations for reducing the risk.
11. Handle unsupported or invalid input.
12. Protect the user's content by minimizing unnecessary storage and processing.

2.3 User Characteristics

PrivacyGuard is mainly intended for:

- Students
- Young social-media users
- People who frequently share images or screenshots online
- Users who share documents or text through online platforms
- General users who want to check their content before publishing it

The application is intended to be simple enough for users without technical or cybersecurity knowledge.

2.4 Constraints

The project has the following constraints:

- The application must be developed for Android.
- Java must be the main programming language.
- The project is developed by a team of three students.
- The project duration is approximately 3–4 months.
- The team has basic knowledge of Java and Python but must learn some Android development concepts.
- The application should work without requiring special hardware.
- The project has limited time and resources, so the detection system will focus on selected types of personal information.
- Detection may not always be perfect and can produce false positives or false negatives.

2.5 Assumptions and Dependencies

The project assumes that:

- The user has an Android device or Android emulator.
- The application has the necessary permissions to access selected files or images.
- OCR and other required libraries are available and compatible with the Android application.
- The user provides content that the application supports.
- Some analysis functions may depend on the quality and readability of the provided content.
- Image metadata can only be detected when such metadata is actually present in the image.

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3. Specific Requirements

3.1 Functional Requirements

3.1.1 Content Input

ID	Priority	Requirement
FR-M-01	Must	The system shall allow the user to enter text for privacy analysis.
FR-M-02	Must	The system shall allow the user to select an image from the device.
FR-M-03	Must	The system shall allow the user to provide a screenshot for analysis.
FR-S-01	Should	The system shall allow the user to take a photo using the device camera.
FR-C-01	Could	The system could support additional document formats in a future version.

3.1.2 Personal Information Detection

ID	Priority	Requirement
FR-M-04	Must	The system shall detect potential phone numbers in the analyzed content.
FR-M-05	Must	The system shall detect potential email addresses in the analyzed content.
FR-M-06	Must	The system shall detect potential addresses or location information when identifiable.
FR-M-07	Must	The system shall detect potential school or workplace information when identifiable.
FR-S-02	Should	The system shall detect potential identification information such as ID or passport numbers when identifiable.
FR-S-03	Should	The system shall detect potential travel or real-time location information when identifiable.

3.1.3 Image and OCR Analysis

ID	Priority	Requirement
FR-M-08	Must	The system shall extract readable text from supported images using OCR.
FR-M-09	Must	The system shall analyze OCR-extracted text for potential personal information.
FR-M-10	Must	The system shall analyze available image metadata.
FR-M-11	Must	The system shall detect GPS/location metadata when it is available.
FR-S-04	Should	The system shall identify other relevant metadata that may create a privacy risk.

3.1.4 Risk Assessment

ID	Priority	Requirement
FR-M-12	Must	The system shall identify the privacy risks detected in the analyzed content.
FR-M-13	Must	The system shall assign an overall privacy risk level.
FR-M-14	Must	The system shall classify the overall risk as Low, Medium, or High.
FR-M-15	Must	The system shall display the information that caused each detected risk.
FR-S-05	Should	The system shall provide a numerical risk score in addition to the risk level.

3.1.5 Results and Recommendations

ID	Priority	Requirement
FR-M-16	Must	The system shall display the analysis results to the user.
FR-M-17	Must	The system shall explain why detected information may represent a privacy risk.
FR-M-18	Must	The system shall provide recommendations for reducing identified privacy risks.
FR-S-06	Should	The system shall identify which detected information should receive the highest attention.
FR-C-02	Could	The system could provide options to blur or hide selected sensitive information.

3.1.6 Error Handling

ID	Priority	Requirement
FR-M-19	Must	The system shall display an appropriate message when the selected content cannot be analyzed.
FR-M-20	Must	The system shall inform the user when required permissions are not available.
FR-S-07	Should	The system shall inform the user when the quality of an image is too low for reliable OCR analysis.

3.2 Non-Functional Requirements

3.2.1 Usability

ID	Priority	Requirement
NFR-M-01	Must	The system shall provide a simple interface that can be used without technical knowledge.
NFR-M-02	Must	The system shall present analysis results in a clear and understandable way.
NFR-S-01	Should	The system should use clear visual indicators for Low, Medium, and High risk levels.

3.2.2 Performance

ID	Priority	Requirement
NFR-M-03	Must	The system shall complete normal content analysis within a reasonable amount of time.
NFR-S-02	Should	The system should provide feedback while a long analysis is being performed.

3.2.3 Privacy and Security

ID	Priority	Requirement
NFR-M-04	Must	The system shall avoid permanently storing users' analyzed content unless necessary.
NFR-M-05	Must	The system shall request only the Android permissions necessary for its functions.
NFR-M-06	Must	The system shall minimize unnecessary transmission of users' sensitive content.
NFR-M-07	Must	The system shall protect analysis results from unauthorized access within the application.
NFR-S-03	Should	The system should perform privacy analysis locally on the device whenever possible.

3.2.4 Reliability

ID	Priority	Requirement
NFR-M-08	Must	The system shall not crash when the user provides unsupported or invalid input.
NFR-M-09	Must	The system shall provide a result even when only some supported privacy information can be detected.

3.2.5 Compatibility

ID	Priority	Requirement
NFR-M-10	Must	The system shall run on supported Android devices.
NFR-S-04	Should	The system should support different common Android screen sizes.

3.2.6 Maintainability

ID	Priority	Requirement
NFR-M-11	Must	The application code shall be organized into understandable components.
NFR-M-12	Must	The application shall use clear names and basic comments where necessary.
NFR-S-05	Should	The detection rules should be designed so that new types of personal information can be added later.

3.2.7 Detection Quality

ID	Priority	Requirement
NFR-M-13	Must	The system shall be tested using representative examples of sensitive and non-sensitive content.
NFR-M-14	Must	The project shall evaluate false positives and false negatives in the detection process.
NFR-S-06	Should	The system should provide sufficiently accurate results for the selected types of personal information.

3.3 MoSCoW Priorities

The requirements are prioritized using the MoSCoW method.

Must Have

These features are necessary for the first working version:

- Android application developed mainly in Java
- Text input
- Image and screenshot input
- Personal information detection
- OCR for images
- Basic image metadata analysis
- GPS metadata detection
- Risk identification
- Low/Medium/High risk classification
- Explanation of detected risks
- Recommendations
- Basic error handling
- Privacy-conscious processing

Should Have

These features are important but are not essential if time becomes limited:

- Camera input
- Detection of identification information
- Detection of travel/location-related information
- Numerical risk score
- Better metadata analysis
- Improved result visualization
- More detailed error messages

Could Have

These features can be implemented if enough time remains:

- Automatic blurring of sensitive information
- Additional document formats
- More advanced detection techniques
- Additional types of privacy risks

Won't Have in the Current Version

The following features are excluded from the current project:

- Direct social-media integration
- Automatic posting to social media
- Continuous monitoring of the user's device
- Direct monitoring of social-media accounts
- Face recognition
- Complete video analysis
- Training a large AI model from scratch

Appendix A – Out of Scope Features

The following features are outside the scope of the current PrivacyGuard project:

1. **Social-media integration** – the application will not connect directly to social-media accounts.
2. **Automatic publishing** – the application will not publish content on behalf of users.
3. **Continuous monitoring** – the application will not continuously monitor files, applications, or device activity.
4. **Face recognition** – the application will not identify people in images.
5. **Complete video analysis** – video processing is not part of the main version.
6. **Large-scale AI model training** – the team will not train a large AI model from scratch.
